Equations of Lines in 3D: Cartesian Equations

Questions 1–20

Question 1. Find the cartesian equation of the line that:

a) passes through $(4, 1, -2)$ and is parallel to $2\mathbf{i} + 3\mathbf{j} + 8\mathbf{k}$
b) passes through $(2, -8, -5)$ and is parallel to $9\mathbf{i} - \mathbf{j} - 2\mathbf{k}$
c) passes through $(-3, -5, 6)$ and is parallel to $2\mathbf{i} - \mathbf{j} - 5\mathbf{k}$
d) passes through $(1, -1, 0)$ and is parallel to $-\mathbf{i} + 3\mathbf{j} + 10\mathbf{k}$
e) passes through $(0, 8, -3)$ and is parallel to $-\mathbf{i} - 5\mathbf{j} + 9\mathbf{k}$
f) passes through $\left(\frac{1}{2}, -3, -\frac{1}{3}\right)$ and is parallel to $5\mathbf{i} - 2\mathbf{j} - 10\mathbf{k}$
g) passes through $\left(-\frac{1}{4}, \frac{1}{2}, \frac{2}{3}\right)$ and is parallel to $10\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$
h) passes through $(2, -5, -7)$ and is parallel to $3\mathbf{i} + 4\mathbf{j}$
i) passes through $(-7, 3, 12)$ and is parallel to $8\mathbf{i} - 3\mathbf{k}$
j) passes through $(-4, 3, 3)$ and is parallel to $-5\mathbf{j} + 6\mathbf{k}$
k) passes through $(-8, 3, -7)$ and is parallel to $\mathbf{i}$
l) passes through $\left(\frac{3}{4}, -\frac{5}{3}, \frac{9}{4}\right)$ and is parallel to $-16\mathbf{i} + 3\mathbf{k}$

Question 2. Find the coordinates of the point when $z = 6$ on the line with equation

$$\frac{x-1}{3} = \frac{y+2}{4} = \frac{z-5}{2}.$$

Question 3. Find the coordinates of the point when $y = 8$ on the line with equation

$$\frac{x+5}{6} = \frac{2-y}{10} = \frac{z+8}{-3}.$$

Question 4. Show that the point $(-5, 20, 6)$ lies on the line with equation

$$\frac{x-3}{-2} = \frac{y-4}{4} = \frac{z+6}{3}.$$

Question 5. Determine whether the point $(2, -8, 3)$ lies on the line with equation

$$\frac{x-9}{3} = \frac{y-13}{9}, \quad z = 3.$$

Question 6. Determine whether the point $(15, 11, -10)$ lies on the line with equation

$$\frac{x+2}{8} = \frac{y+57}{32} = \frac{z+18}{4}.$$

Question 7. Find a cartesian equation of the line passing through the following pairs of points:

a) $(0, 0, 0)$ and $(2, 3, 6)$
b) $(1, 2, -4)$ and $(3, 5, -7)$
c) $(4, 0, -8)$ and $(5, 1, 0)$
d) $(2, 5, -1)$ and $(2, -3, 2)$
e) $(3, 8, 1)$ and $(9, 8, -6)$
f) $(5, -3, -2)$ and $(6, -7, -2)$

Question 8. Determine whether the following pair of lines are parallel or not:

$$l_1 : \frac{2x-3}{6} = \frac{5y+11}{-4} = \frac{8z-9}{12}, \qquad l_2 : \frac{1-3x}{9} = \frac{10y-13}{8} = \frac{5-2z}{3}.$$

Question 9. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \frac{x-3}{2} = \frac{y+2}{3} = \frac{z-74}{10}, \qquad l_2 : \frac{x+1}{3} = \frac{y-4}{6} = \frac{z+2}{8}.$$

Question 10. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \frac{5x+25}{-8} = \frac{5y-25}{2} = \frac{5z+3}{-4}, \qquad l_2 : \frac{5x+20}{-3} = \frac{5y+25}{-3} = \frac{5z-45}{2}.$$

Question 11. Determine whether the following pair of lines are coincident, parallel, skew or if they intersect. If they intersect, find the coordinates of the point of intersection.

a) $l_1 : \frac{x+1}{4} = \frac{y-3}{-5} = \frac{2z-7}{-4}$, $l_2 : \frac{x-9}{-4} = \frac{y+2}{5} = \frac{3z+1}{6}$

b) $l_1 : \frac{x+2}{3} = \frac{4y-1}{3} = 2 - z$, $l_2 : \frac{x+3}{3} = \frac{8y+3}{6} = 3 + z$

c) $l_1 : \frac{4x-1}{2} = y + 7 = \frac{3-z}{8}$, $l_2 : \frac{6x-7}{3} = \frac{2y+1}{4} = \frac{2z-3}{2}$

d) $l_1 : \frac{1-4x}{8} = \frac{y+5}{3} = \frac{5z-65}{4}$, $l_2 : \frac{3x-2}{5} = \frac{7-4y}{9} = 3 - z$

Question 12. It is given that the following lines intersect:

$$l_1 : \frac{x-6}{-3} = \frac{y+1}{2} = \frac{z-7}{4}, \qquad l_2 : \frac{x-15}{k} = \frac{y+3}{9} = \frac{z+12}{6},$$

where k is a constant.

a) Find the value of k.

b) Find the position vector of the point of intersection of the two lines.

Question 13. Find the acute angle between each pair of lines:

a) $l_1 : \frac{x-3}{2} = \frac{y+7}{9} = \frac{z-3}{4}, \quad l_2 : \frac{x+7}{4} = \frac{y-1}{2} = \frac{z+8}{3}$

b) $l_1 : \frac{x+2}{-8} = \frac{y-1}{2} = z + 2, \quad l_2 : \frac{x-3}{4} = \frac{y+2}{3} = \frac{z-4}{2}$

c) $l_1 : \frac{3-x}{2} = \frac{y+1}{-3} = 3 - z, \quad l_2 : x + 2 = 3 - y = 4 - z$

d) $l_1 : x = 3, \quad \frac{2-y}{2} = \frac{z+8}{1}, \quad l_2 : x = 8, \quad y - 9 = \frac{z-4}{5}$

e) $l_1 : \frac{2x-3}{4} = \frac{7-3y}{6} = \frac{2z+1}{4}, \quad l_2 : \frac{2x+1}{3} = \frac{6-4y}{5} = 10 - z$

f) $l_1 : 5 - x = \frac{y-9}{4}, \quad z = 5, \quad l_2 : \frac{8x+3}{3} = \frac{4-z}{5}, \quad y = -2$

Question 14. Two small intergalactic asteroids are travelling on different straight-line paths, which can be modelled by the equations:

$$l_1 : \frac{x - 2140}{10} = \frac{y - 425}{9}, \quad z = 670, \qquad l_2 : \frac{x + 200}{15k} = \frac{y - 380}{18} = \frac{475 - z}{5},$$

where k is a constant. The two lines intersect.

a) Find the value of k.

b) Find the coordinates of where the two lines intersect.

c) It is given that at time T seconds, the asteroids are both the same distance from where their paths intersect. Explain why it is not necessarily the case that the asteroids will impact one another.

Question 15. The line l_1 has the equation:

$$\frac{x+1}{4} = 3 - y = \frac{3z+4}{6}$$

and l_2 passes through $(5, 0, 4)$ and $(6, k, k+1)$, where k is a constant. Given that l_1 and l_2 are perpendicular,

a) find the value of k

It is given that l_3 passes through $(5, 0, 4)$ and $(6, k, k-1)$ and that θ is the angle between l_1 and l_3.

a) Show that $\cos\theta = -\frac{2\sqrt{6}}{21}$.

Question 16. Two lines are given by

$$l_1 : \frac{5-x}{2} = 3 - y = 2z, \qquad l_2 : \frac{x-2}{8-k} = \frac{y+5}{2p+5} = \frac{z-6}{-4}.$$

Find the values of k and p given that:

a) l_1 and l_2 are parallel

b) l_1 and l_2 intersect and are perpendicular

Question 17. The plane Π contains the following two lines:

$$l_1 : \frac{x-3}{7} = \frac{2y+1}{2} = 1-z, \qquad l_2 : \frac{x-8}{3} = \frac{2y-21}{6} = \frac{2z+8}{-3}.$$

Find a vector equation of the plane Π.

Question 18. The plane Π contains the following two lines:

$$l_1 : 3-x = \frac{y-4}{-5} = z+1, \qquad l_2 : \frac{x+2}{2} = 2-y = \frac{z+20}{6}.$$

Find a cartesian equation of the plane Π.

Question 19. Write the following cartesian equations of lines in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$:

a) $\frac{x+1}{2} = \frac{y-1}{3} = \frac{z+2}{-3}$

b) $\frac{x}{8} = \frac{y+9}{4} = \frac{z-1}{-5}$

c) $\frac{x+8}{3} = \frac{2-y}{3} = \frac{z}{10}$

d) $\frac{7-x}{3} = \frac{y-8}{12} = z + 3$

e) $\frac{2x-3}{5} = \frac{3y+1}{2} = \frac{5z-3}{2}$

f) $\frac{3-4x}{6} = \frac{5-2y}{2} = \frac{6-8z}{-5}$

g) $x = -4, \quad \frac{y-3}{-4} = \frac{z+12}{7}$

h) $x - 9 = \frac{2y-4}{-5}, \quad z = 1$

i) $x = 6 - z, \quad y = 5$

Question 20. Write the following cartesian equations of lines in vector product form:

a) $\frac{x-3}{5} = \frac{y-2}{2} = \frac{z-4}{6}$

b) $\frac{x}{9} = \frac{2-y}{3} = 5 - z$

c) $3 - x = \frac{y+1}{2} = 3 - z$

d) $\frac{3x+2}{5} = 6 - y, \quad z = 2$

e) $x = \frac{1}{3}, \quad \frac{4y-3}{2} = \frac{5-3z}{5}$

f) $x = 4, \quad 3y = 4 - 2z$